Applied Linear Algebra in Data Analysis

Matrix Inverses

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Rank of a matrix A

- Column Rank is always equal to the row rank.
- Rank tells us the number of independent columns/rows in the matrix.
- **Full rank matrix** $\mathbf{A}$: $\text{rank}(\mathbf{A}) = \min(p, q)$
- **Rank deficient matrix** $\mathbf{A}$: $\text{rank}(\mathbf{A}) < \min(p, q)$

Matrix Inverse

Consider the square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$. $\mathbf{B} \in \mathbb{R}^{n \times n}$ is the inverse of $\mathbf{A}$, if $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$, and $\mathbf{B}$ is represented as $\mathbf{A}^{-1}$.

- Not all matrices have inverses. A matrix with an inverse is called **non-singular**, otherwise it is called **singular**.
- For a non-singular matrix $\mathbf{A}$, $\mathbf{A}^{-1}$ is unique. $\mathbf{A}^{-1}$ is both the left and right inverse.
- A matrix $\mathbf{A}$ has an inverse, if and only if $\mathbf{A}$ is full rank, i.e. $rank\,(\mathbf{A}) = n$
- $\mathbf{Ax} = \mathbf{b}$ can be solved as follows, $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. *It is never solved like this in practice.*
- Inverse of product of matrices, $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.
- $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$ and $(\mathbf{A}^{-1})^{\top} = (\mathbf{A}^{\top})^{-1}$

Matrix Inverse

Some interesting observations about matrix inverses:

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \qquad \mathbf{A}^{-1} = \begin{bmatrix} \tilde{\mathbf{b}}_1^\top \\ \tilde{\mathbf{b}}_2^\top \\ \vdots \\ \mathbf{b}_n^\top \end{bmatrix} \qquad \mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$$

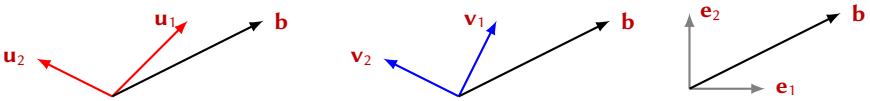
$$\implies \tilde{\mathbf{b}}_i^\top \mathbf{a}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \quad \text{and} \quad \tilde{\mathbf{a}}_i^\top \mathbf{b}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$$\mathbf{A}\mathbf{b}_i = \mathbf{e}_i \quad \text{and} \quad \tilde{\mathbf{b}}_i^\top \mathbf{A} = \mathbf{e}_i^\top \implies \begin{bmatrix} \mathbf{A} & | & \mathbf{I}_n \end{bmatrix} \xrightarrow{\text{Gauss-Jordan}} \begin{bmatrix} \mathbf{I}_n & | & \mathbf{A}^{-1} \end{bmatrix}$$

Representation of vectors in a basis

Consider the vector space $\mathbb{R}^n$ with basis $\{\mathbf{v}_1, \mathbf{v}_2, \ldots \mathbf{v}_n\}$. Any vector in $\mathbf{b} \in \mathbb{R}^n$ can be represented as a linear combination of vectors $\mathbf{v}_i$,

$$\mathbf{b} = \sum_{i=1}^{n} \mathbf{v}_i \mathbf{a}_i = \mathbf{V}\mathbf{a}; \quad \mathbf{a} \in \mathbb{R}^n, \quad \mathbf{V} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix} \in \mathbb{R}^{n \times n}$$



$\{\mathbf{v}_1, \mathbf{v}_2\}$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ and $\{\mathbf{e}_1, \mathbf{e}_2\}$ are valid basis for $\mathbb{R}^2$, and the presentation for $\mathbf{b}$ in each one of them is different.

Matrix Inverse

Consider the equation $\mathbf{A}\mathbf{x} = \mathbf{y}$, where $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Let us assume $\mathbf{A}$ is non-singular $\implies$ columns of $\mathbf{A}$ represent a basis for $\mathbb{R}^n$.

What does $\mathbf{x}$ represent? It is the representation of $\mathbf{y}$ in the basis consisting of the columns of $\mathbf{A}$.

$$\mathbf{y} = \mathbf{A}\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \sum_{i=1}^{n} \mathbf{a}_i x_i$$

$$\implies \quad \mathbf{x} = \mathbf{A}^{-1}\mathbf{y} = \begin{bmatrix} \tilde{\mathbf{b}}_1^\top \\ \tilde{\mathbf{b}}_2^\top \\ \vdots \\ \tilde{\mathbf{b}}_n^\top \end{bmatrix} \mathbf{y} = \begin{bmatrix} \tilde{\mathbf{b}}_1^\top \mathbf{y} \\ \tilde{\mathbf{b}}_2^\top \mathbf{y} \\ \vdots \\ \tilde{\mathbf{b}}_n^\top \mathbf{y} \end{bmatrix}$$

$\mathbf{A}^{-1}$ is a matrix that allows change of basis to the columns of $\mathbf{A}$ from the standard basis!

Left Inverse

- Consider a rectangular matrix $\mathbf{A} \in \mathbb{R}^{p \times q}$. There exists no inverse $\mathbf{A}^{-1}$ for this matrix.
- But, there exist two matrices $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{q \times p}$, such that,

$$\mathbf{CA} = \mathbf{I}_q \quad \text{or} \quad \mathbf{AB} = \mathbf{I}_p$$

- Both cannot be true for a rectangular matrix, only one can be true when the matrix is full rank.
- A rectangular matrix can only have either a left or a right inverse.

Left Inverse

- Any non-zero $\mathbf{a} \in \mathbb{R}^{n\times 1}$ is left invertible: $\mathbf{ba} = 1, \quad \mathbf{b} \in \mathbb{R}^{1\times n}; \quad \mathbf{b}^T = \frac{\mathbf{a}}{\|\mathbf{a}\|^2} + \alpha \mathbf{a}^{\perp}$

- This can be generalized to $\mathbf{A} \in \mathbb{R}^{p\times q}$, $m > n$.

$$\left(\mathbf{C} + \hat{\mathbf{C}}\right)\mathbf{A} = \mathbf{I}_m \text{ where } \mathbf{C}, \hat{\mathbf{C}} \in \mathbb{R}^{n\times m}, \quad \hat{\mathbf{C}}\mathbf{A} = \mathbf{0}$$

- Condition for left inverse of $\mathbf{A}$ to exist: *Colmuns of* $\mathbf{A}$ *must be independent.*
$\longrightarrow$ *rank* $(\mathbf{A}) = n \longrightarrow \mathbf{Ax} = 0 \implies \mathbf{x} = 0$.

- $\mathbf{Ax} = \mathbf{b}$ can be solved, if and only if $\mathbf{A}\left(\mathbf{Cb}\right) = \mathbf{b}$, where $\mathbf{CA} = \mathbf{I}_n$.

Left Inverse:Examples

Let's find the left inverse of the following matrices:

- $A = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

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Let's find the left inverse of the following matrices:

- $\mathbf{A} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ $\qquad \mathbf{C} = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_1 + 2\alpha_2 - 1 \end{bmatrix}.$

- $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{bmatrix}$

Left Inverse:Examples

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- $\mathbf{A} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ $\qquad \mathbf{C} = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_1 + 2\alpha_2 - 1 \end{bmatrix}.$

- $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{bmatrix}$ $\qquad \mathbf{C} = \begin{bmatrix} \frac{1}{2} - \frac{3}{2}\alpha_1 & \frac{1}{2} - \frac{1}{2}\alpha_1 & \alpha_1 \\ \frac{1}{2} - \frac{3}{2}\alpha_2 & -\frac{1}{2} - \frac{1}{2}\alpha_2 & \alpha_2 \end{bmatrix}$

Right Inverse

- For $\mathbf{A} \in \mathbb{R}^{p \times q}$, $n > m$ with full rank, $\mathbf{AB} = \mathbf{I}_m \longrightarrow \mathbf{B}$ is the right inverse.
- Right inverse of $\mathbf{A}$ exists only if the rows of $\mathbf{A}$ are independent, i.e. $rank\left(\mathbf{A}\right) = m$ $\longrightarrow \mathbf{A}^T\mathbf{x} = \mathbf{0} \implies \mathbf{x} = \mathbf{0}$
- $\mathbf{Ax} = \mathbf{b}$ can be solved for any $\mathbf{b}$. $\mathbf{x} = \mathbf{Bb} \implies \mathbf{A}\left(\mathbf{Bb}\right) = \mathbf{b}$.
- There are an infitnite number of $\mathbf{B}$s $\implies$ an infinite number of solutions $\mathbf{x}$.

Right Inverse:Examples

Let's find the right inverse of the following matrices:

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Right Inverse:Examples

Let's find the right inverse of the following matrices:

- $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$ $\qquad$ $\mathbf{B} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ 1 - \alpha_1 - \alpha_2 \end{bmatrix}$.

- $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$

Right Inverse:Examples

Let's find the right inverse of the following matrices:

- $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$ $\qquad \mathbf{B} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ 1 - \alpha_1 - \alpha_2 \end{bmatrix}.$

- $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$ $\qquad \mathbf{C} = \begin{bmatrix} \frac{1}{2} - \alpha_1 & \frac{1}{2} - \alpha_2 \\ \frac{1}{2} & -\frac{1}{2} \\ \alpha_1 & \alpha_2 \end{bmatrix}$

Pseudo Inverse

- Consider a tall matrix $\mathbf{A} \in \mathbb{R}^{p \times q}$ with independent columns. It turns out the Gram matrix $\mathbf{A}^\top \mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible. If that is the case then,

$$\left(\mathbf{A}^\top \mathbf{A}\right)^{-1} \mathbf{A}^\top \mathbf{A} = \mathbf{I}_n; \quad \left(\mathbf{A}^\top \mathbf{A}\right)^{-1} \mathbf{A}^\top \text{ is a left inverse.}$$

- $\mathbf{A}^\dagger = \left(\mathbf{A}^\top \mathbf{A}\right)^{-1} \mathbf{A}^\top$ is called the *pseudo inverse* or the *Moore-Penrose inverse*.

- For the case of a fat, wide matrix, we have $\mathbf{A}^\dagger = \mathbf{A}^\top \left(\mathbf{A}\mathbf{A}^\top\right)^{-1}$.

- When $\mathbf{A}$ is square and invertible, $\mathbf{A}^\dagger = \mathbf{A}^{-1}$.

Pseudo Inverse: Examples

Find the pseudo inverse of $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$.

Pseudo Inverse: Examples

Find the pseudo inverse of $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$.

Compute the matrix $\mathbf{A}\mathbf{A}^{\dagger}$ and $\mathbf{A}^{\dagger}\mathbf{A}$.

Pseudo Inverse: Examples

Find the pseudo inverse of $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$.

Compute the matrix $\mathbf{A}\mathbf{A}^{\dagger}$ and $\mathbf{A}^{\dagger}\mathbf{A}$.

Find the QR decompositon of $\mathbf{A}$ and compute $\mathbf{Q}\mathbf{Q}^{\top}$.

Pseudo Inverse: Examples

Solve $\mathbf{Ax} = \mathbf{b}$ using the $\mathbf{A}^\dagger$. $\mathbf{A} = \begin{bmatrix} 1 & 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} 3 \end{bmatrix}$. Find $\mathbf{x}$.

Write down all the possible solution $\mathbf{x}$. What is the $\mathbf{x}$ with the smallest length? What is $\mathbf{A}^\dagger\mathbf{b}$?

Matrix Inverse and Pseudo Inverse through QR factorization

- Consider an invertible, square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$.

$$\mathbf{A} = \mathbf{QR} \implies \mathbf{A}^{-1} = (\mathbf{QR})^{-1} = \mathbf{R}^{-1}\mathbf{Q}^{-1} = \mathbf{R}^{-1}\mathbf{Q}^{\top}$$

  where, $\mathbf{R}, \mathbf{Q} \in \mathbb{R}^{n \times n}$. $\mathbf{R}$ is upper triangular, and $\mathbf{Q}$ is an orthogonal matrix.

- In the case of a left invertible rectangular matrix $\mathbf{A} \in \mathbb{R}^{p \times q}$, we can factorize $\mathbf{A} = \mathbf{QR}$, with $\mathbf{Q} \in \mathbb{R}^{p \times q}$ and $\mathbf{R} \in \mathbb{R}^{n \times n}$.

$$\mathbf{A}^{\dagger} = \left(\mathbf{A}^{\top}\mathbf{A}\right)^{-1}\mathbf{A}^{\top} = \left(\mathbf{R}^{\top}\mathbf{Q}^{\top}\mathbf{QR}\right)^{-1}\mathbf{R}^{\top}\mathbf{Q}^{\top} = \left(\mathbf{R}^{\top}\mathbf{R}\right)^{-1}\mathbf{R}^{\top}\mathbf{Q}^{\top} = \mathbf{R}^{-1}\mathbf{Q}^{\top}$$

Matrix Inverse and Pseudo Inverse through QR factorization

- For a right invertible wide, fat matrix, we can find out the pseudo-inverse of $\mathbf{A}^\top$, and then take the transpose of the pseudo-inverse.

$$\mathbf{A}\mathbf{A}^\dagger = \mathbf{I} \implies \left(\mathbf{A}^\dagger\right)^\top \mathbf{A}^\top = \left(\mathbf{A}^\top\right)^\dagger \mathbf{A}^\top = \mathbf{I}$$

$$\mathbf{A}^\top = \mathbf{Q}\mathbf{R} \implies \left(\mathbf{A}^\top\right)^\dagger = \mathbf{R}^{-1}\mathbf{Q}^\top = \left(\mathbf{A}^\dagger\right)^\top \implies \mathbf{A}^\dagger = \mathbf{Q}\mathbf{R}^{-T}$$